Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · STATIONARITY TESTS (ADF, KPSS)

is the series stationary?

Table. Stationarity tests

Test	Statistic	Lag	p	Conclusion
ADF	−3.83	4	.020	stationary
KPSS	0.08	4	> .100	stationary
PP	−13.17	4	.354	non-stationary

ADF and KPSS have opposite null hypotheses - the conclusion column reconciles them. The Statistic column holds two different metrics (ADF: Dickey–Fuller τ; KPSS: LM statistic); KPSS p-values are interpolated and shown bounded (e.g. "> .10" / "< .01") rather than exact.

Figure. Series & structure

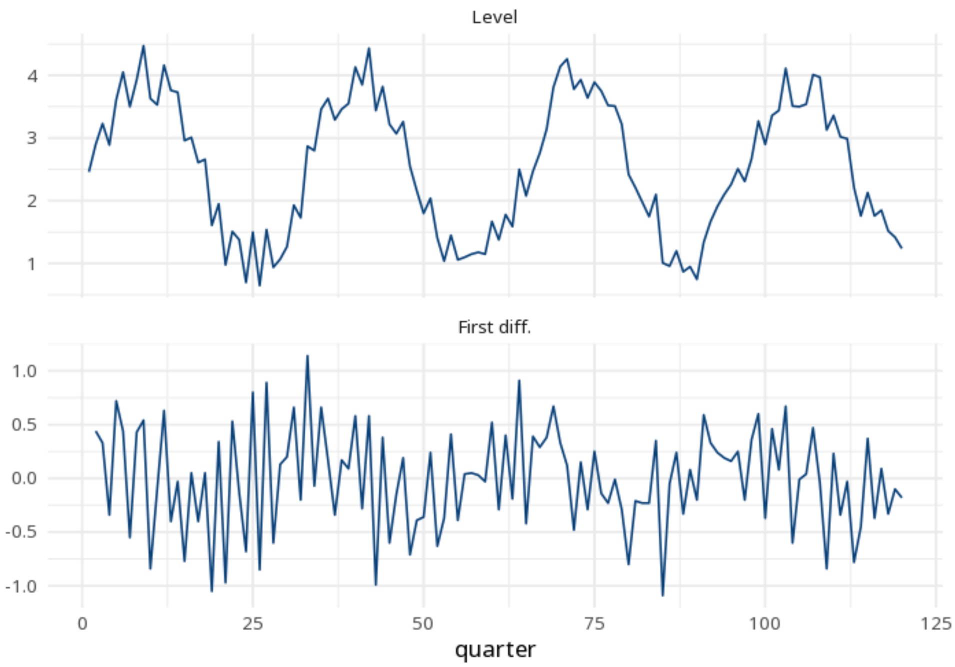
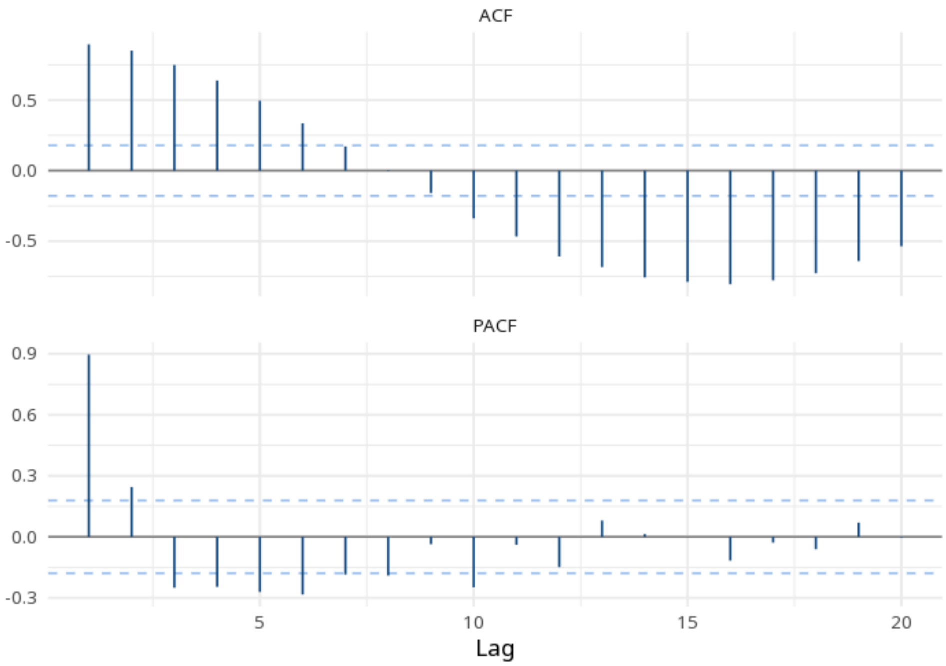


Figure. Series & structure



Understanding the numbers

Test. Which stationarity test this row reports - ADF, KPSS, or Phillips–Perron. Here, the table reports: ADF, KPSS, PP.

Statistic. The test statistic itself - the Dickey–Fuller τ for ADF, the LM statistic for KPSS, or the Phillips–Perron Z for PP (not on the same scale). Here, the ADF statistic is $\tau = -3.83$.

Lag. The number of lags the test used to account for autocorrelation. Here, ADF used lag = 4.

p. For ADF, a small p means stationary; for KPSS, a small p means non-stationary - the tests have opposite null hypotheses. Here, ADF gives p .020.

Conclusion. The verdict reconciling the (opposite-null) tests for this row. Here, the ADF row's conclusion is "stationary".

How to read this test

For ADF, a small p means stationary; for KPSS, a small p means non-stationary. If a series is non-stationary, difference it before modeling. The conclusion column states the verdict. Failing to reject is not proof of the null for either test (both have low power), so strengthen the verdict by also reading the ACF rather than relying on a single p-value. Method: R tseries package ADF/KPSS/PP (Dickey & Fuller; Kwiatkowski, Phillips, Schmidt & Shin, 1992; Phillips & Perron, 1988). Your significance threshold (α) is 0.05. Tests run: ADF, KPSS, and Phillips–Perron.

APA template: ADF gave $\tau=-3.83$, $p = .020$; KPSS gave LM=0.08, $p > .100$; Phillips–Perron gave $Z=-13.17$, $p = .354$ ($\alpha=0.05$).

Statistical basis: Augmented Dickey-Fuller unit-root test (Dickey, D. A., Fuller, W. A. (1979). "Distribution of the estimators for autoregressive time series with a unit root." Journal of the American Statistical Association, 74(366a), 427-431.); KPSS stationarity test (Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., Shin, Y. (1992). "Testing the null hypothesis of stationarity against the alternative of a unit root." Journal of Econometrics, 54(1-3), 159-178.); Phillips-Perron unit-root test (Phillips, P. C. B., Perron, P. (1988). "Testing for a unit root in time series regression." Biometrika, 75(2), 335-346.); Implementation (tseries package) (Trapletti A, Hornik K (2024). tseries: Time Series Analysis and Computational Finance. R package.). Full references in the exported CITATIONS.txt.

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